Electronic Retail Sales Analysis

1. Project Overview

This project examines retail customer shopping behaviour by analyzing transactional data from 5,200 sales across different product brands. The aim is to uncover insights into customer segments, product preferences, sales performance, profitability, and overall business performance, which will support strategic business decisions.

2. Business Introduction

An electronics retail company sells a variety of electronic products and accessories across multiple regions. The company works with well-known brands such as Dell, Asus, HP, Samsung, Lenovo, and Apple.

Despite generating substantial sales in electronics, management lacks an integrated system to identify which customers, products, brands, and regions contribute the most to revenue and profit. This project analyzes business performance using sales, customer, and product data, from data cleaning to creating an interactive dashboard. The goal is to optimize customer behaviour and retention while solidifying the key drivers of revenue and profit.

3. Project Goal

The dashboard helps the electronic retail company management team monitor electronic sales performance across regions and product categories and provide insights into revenue, profitability, monthly customer trends, and product performance to support business decisions.

4. Data Cleaning and Transformation

Tools: Power Query (Power BI)

Before analysis, I reviewed and cleaned the dataset in **Power Quer**y to ensure consistency and accuracy. The dataset comprised three main tables: **Sales, Customers, and Products**. The following data-cleaning steps were taken:

1. **Checked for missing values and duplicate rows:** The dataset was reviewed for duplicate records and missing values using the filter dropdown method, but none were found.

2. **Standardized text values:** In the Region column, I standardized “north” to “North,” and in the Gender column, I changed “female” to “Female.”

3. **Formatted product categories:** I identified inconsistencies in the product category column, where some products were incorrectly classified. For instance, items that should belong to Electronics were sometimes categorized as Accessories and vice versa. To address this, I created a new, corrected category column to ensure consistent product classification before analysis.

4. **Formatted dates:** The sales date and customer signup date columns were reviewed and converted into appropriate date formats.

5. Data Modelling

Tools: Power BI Data Model, DAX

1. Defined relationships between tables (e.g products, sales, customers).
2. Created DAX measures for key KPIs, including total customers, total sales, total profit, and total revenue.

6. Data Analysis using SQL

We performed structured analysis in Microsoft SQL to answer key business questions:

1. Profit by Brand - Compared total profit generated by different brands purchased by customers.

Results Messages

	Brand	Total_Profit	Profit_Contribution_Percentage
1	Apple	196890	21.130000000000
2	Lenovo	160140	17.180000000000
3	Samsung	159510	17.120000000000
4	HP	147990	15.880000000000
5	Asus	141525	15.190000000000
6	Dell	125850	13.500000000000

2. Sales Volume by Colour - Identified product colours that were purchased most by customers.

Results Messages

	Color	Total_Transactions
1	White	2051
2	Gray	1744
3	Black	1405

3. Sales by Product Category - Identified product categories that were purchased most by customers.

Results Messages

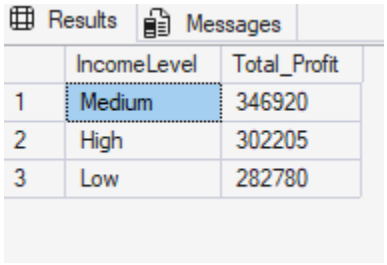
	Category	Total_Transactions	Total_Units_Sold
1	Electronics	2964	8896
2	Accessories	2236	6761

4. Revenue by Region - Calculated total revenue contribution of each region.

Results Messages

	Region	Total_Revenue
1	North	963300
2	East	796700
3	South	703000
4	West	643350

5. Profit by Income Level - Compared total profit generated by the customer's different income brackets.



Results | Messages

	IncomeLevel	Total_Profit
1	Medium	346920
2	High	302205
3	Low	282780

7. Dashboard in Power BI

Tools: Power BI Interactive Dashboard features

1. Built an interactive dashboard with filters, slicers, and drill-through functionality.
2. Used bar charts, line graphs, donut charts, and matrices to visualize customer trends and product profitability.
3. Created KPIs to track our project goal and compared it to the previous month's progress.

8. Key Insights

1. The business generated over ₦3.1 million in revenue.

2. The business generated approximately ₦931,900 in profit, with Apple contributing the highest share at about 21.13% of the total profit.

3. A total of 5,200 sales transactions were recorded, with white products being the most frequently sold at 2,100 sales, while black products had the lowest sales volume.

4. Customers with medium incomes generated the highest profits, making them the most profitable segment.

5. Customer sign-up records occur on dates in May 2020, June 2021, and July 2022. Since these dates span different years, the available data does not establish a continuous monthly increase or decline in customer acquisition.

9. Recommendations

1. **Prioritize profitable brands -** Management should evaluate Apple's sales consistency, stock availability, and product-level margins before increasing inventory commitments.

2. **Develop targeted offers for medium-income customers -** Analyze this segment preferred brands, product categories, and average transaction value to create relevant promotions and encourage repeat purchases.

3. **Expand market based on regional performance -** Focus marketing and operations efforts in the North and investigate what drives its performance.

4. **Improve inventory planning using colour and product demand -** Avoid discontinuing slower-selling products until their margins and customer demand have been evaluated.

5. **Improve customer acquisition trackin**g - Collect consistent monthly customer sign up data.